PROBLEM 01

Rocket Helical Trajectory

Original statement · Problems.pdf, p. 6

MIT OCW Benchmark

An out-of-control rocket traces an upward helical trajectory described in Cartesian coordinates for $t > 0$ by:

$$x = a\cos(\omega t) \tag{1}$$

$$y = a\sin(\omega t) \tag{2}$$

$$z = v_z t \tag{3}$$

where a, v_z, and ω are constants.

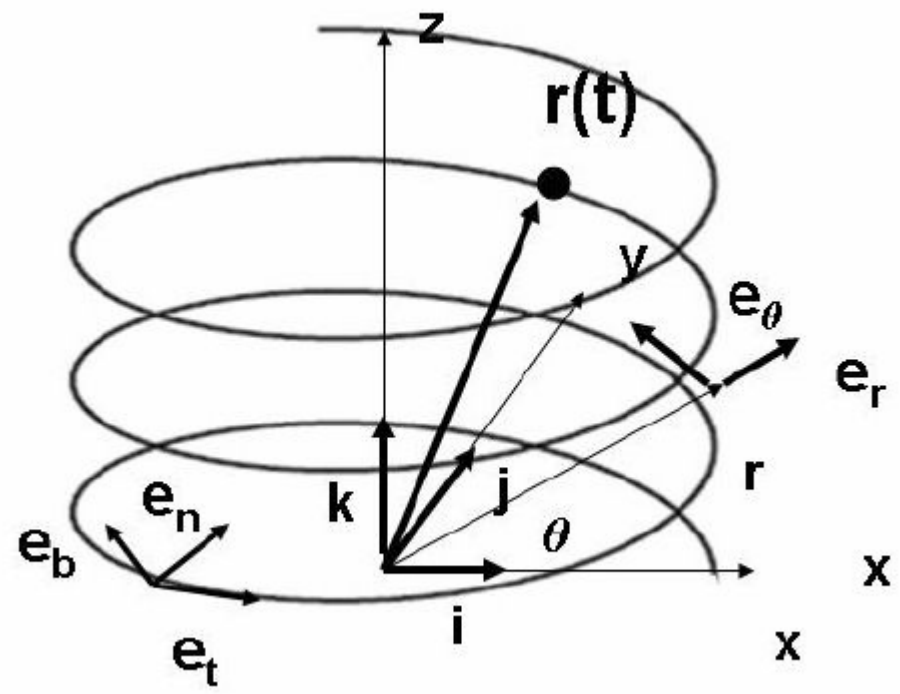


Figure: Helical trajectory with position r(t), the Cartesian basis i, j, k, the cylindrical basis e_R, e_theta, e_z and the intrinsic basis e_t, e_n, e_b (taken from MIT OCW) (Problems.pdf, p. 6).

(a) Express the position, velocity, and acceleration vectors in the Cartesian x, y, z system using unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

(b) Express the position, velocity, and acceleration vectors in the cylindrical coordinate system using unit vectors $\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z$.

(c) Express the velocity and acceleration vectors using s, ρ and the vector basis $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$.

(d) Compute the modulus of the velocity vector.

(e) Express the evolution of the arc-length coordinate as a function of time.

(f) Write down the trajectory as a function of the arc-length coordinate.

(g) Where is the center of curvature for this curve when the particle is located at $(x, y, z) = (a, 0, 0)$?

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** an out-of-control rocket, modelled as a point particle P, ascends along a circular helix around the vertical z-axis.
- **Hypotheses:** $a > 0$, $v_z > 0$ and $\omega > 0$ are constants and $t > 0$.
- **Constraints and degrees of freedom:** the statement gives the three Cartesian coordinates as functions of time, so the particle is on the helix, which satisfies the two holonomic relations $x = a\cos(\omega z / v_z)$ and $y = a\sin(\omega z / v_z)$ (substituting $z = v_z t$ recovers the statement). Two independent relations leave $\mathrm{CDOF} = 3 - 2 = 1$: a single coordinate ($t$, or the arc length s) fixes the position on the helix.
- **Coordinates:** the Cartesian coordinates (x, y, z) of the statement; for the questions we also use the cylindrical coordinates (R, θ, z) and the arc length s.
- **Parameters:** radius a [m], angular frequency ω [rad/s], vertical velocity v_z [m/s], time t [s].

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, fixed to the launch site with $\mathbf{k}_0$ along the vertical z-axis.
- **Cylindrical basis:** $\mathcal{B}_{\text{cyl}} = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z\}$, a rotation of $\mathcal{B}_0$ by the angle $\theta = \omega t$ about $\mathbf{k}_0$:

$$[{}_0R_{\text{cyl}}] = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_{\text{cyl}}] = \cos^2(\omega t) + \sin^2(\omega t) = 1, \qquad [{}_0R_{\text{cyl}}][{}_0R_{\text{cyl}}]^T = I$$

The product RR^T has diagonal entries $\cos^2(\omega t) + \sin^2(\omega t) = 1$ (and 1) and off-diagonal entries $\cos(\omega t)\sin(\omega t) - \sin(\omega t)\cos(\omega t) = 0$.

- The basis vectors, from the columns of the matrix:

$$\mathbf{e}_R = \cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0, \qquad \mathbf{e}_\theta = -\sin(\omega t)\mathbf{i}_0 + \cos(\omega t)\mathbf{j}_0, \qquad \mathbf{e}_z = \mathbf{k}_0$$

- **Time derivatives of the unit vectors** (direct differentiation of their explicit components, with the chain rule $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t)$ and $\frac{d}{dt}\sin(\omega t) = \omega\cos(\omega t)$; Notes Sec. 2.6.2):

$$\frac{d\mathbf{e}_R}{dt} = -\omega\sin(\omega t)\,\mathbf{i}_0 + \omega\cos(\omega t)\,\mathbf{j}_0 = \omega\,\mathbf{e}_\theta, \qquad \frac{d\mathbf{e}_\theta}{dt} = -\omega\cos(\omega t)\,\mathbf{i}_0 - \omega\sin(\omega t)\,\mathbf{j}_0 = -\omega\,\mathbf{e}_R, \qquad \frac{d\mathbf{e}_z}{dt} = \mathbf{0}$$

- **Intrinsic basis:** $\mathcal{B}_F = \{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ (tangent, principal normal toward the centre of curvature, binormal $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$), obtained in 3.1 and related to $\mathcal{B}_{\text{cyl}}$ by a second change-of-basis matrix.

Derivation

3.1 Kinematics

(a) Cartesian position, velocity and acceleration

Pedagogical Justification: the position is given by its Cartesian components in the fixed basis, whose unit vectors are constant ($\dot{\mathbf{i}}_0 = \dot{\mathbf{j}}_0 = \dot{\mathbf{k}}_0 = \mathbf{0}$). Velocity and acceleration follow by differentiating term by term.

$$\mathbf{r}(t) = a\cos(\omega t)\,\mathbf{i}_0 + a\sin(\omega t)\,\mathbf{j}_0 + v_z t\,\mathbf{k}_0$$

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = -a\omega\sin(\omega t)\,\mathbf{i}_0 + a\omega\cos(\omega t)\,\mathbf{j}_0 + v_z\,\mathbf{k}_0$$

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = -a\omega^2\cos(\omega t)\,\mathbf{i}_0 - a\omega^2\sin(\omega t)\,\mathbf{j}_0 + 0\,\mathbf{k}_0 = -a\omega^2\left(\cos(\omega t)\,\mathbf{i}_0 + \sin(\omega t)\,\mathbf{j}_0\right)$$

(b) Cylindrical position, velocity and acceleration

> **Pedagogical Justification:** the helix lies on a cylinder, so $R = \sqrt{x^2 + y^2} = a$ is constant and $\theta = \omega t$. The vectors $\mathbf{e}_R$ and $\mathbf{e}_\theta$ are not constant, so we use the product rule with the derivatives of Phase 2.

$$\mathbf{r} = a\,\mathbf{e}_R + v_z t\,\mathbf{e}_z$$

$$\mathbf{v} = \dot{a}\,\mathbf{e}_R + a\frac{d\mathbf{e}_R}{dt} + \frac{d(v_z t)}{dt}\mathbf{e}_z + v_z t\frac{d\mathbf{e}_z}{dt} = 0 + a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z + 0 = a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z$$

$$\mathbf{a} = \frac{d(a\omega)}{dt}\mathbf{e}_\theta + a\omega\frac{d\mathbf{e}_\theta}{dt} + \frac{dv_z}{dt}\mathbf{e}_z + v_z\frac{d\mathbf{e}_z}{dt} = 0 + a\omega\,(-\omega\,\mathbf{e}_R) + 0 + 0 = -a\omega^2\,\mathbf{e}_R$$

(d) Modulus of the velocity

> **Pedagogical Justification:** the speed is the norm of the velocity vector, the rate at which the distance along the path is travelled. It is computed before part (c) because the intrinsic basis and the radius of curvature need it.

$$v = \|\mathbf{v}\| = \sqrt{a^2\omega^2\left(\sin^2\omega t + \cos^2\omega t\right) + v_z^2} = \sqrt{a^2\omega^2 + v_z^2} \equiv V$$

The speed is constant for all t.

(c) Intrinsic basis, and velocity and acceleration in terms of s and ρ

> **Pedagogical Justification:** the Frenet-Serret basis follows the curve itself (Notes Sec. 2.4): $\mathbf{e}_t$ is the direction of the velocity, $\mathbf{e}_n$ points toward the centre of the osculating circle and $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$ completes a right-handed triad. The intrinsic acceleration is (Notes Eq. 2.17)

$$\mathbf{a} = \ddot{s}\,\mathbf{e}_t + \frac{\dot{s}^2}{\rho}\,\mathbf{e}_n, \qquad \mathbf{v} = \dot{s}\,\mathbf{e}_t$$

- **Tangent.** $\mathbf{e}_t = \dfrac{\mathbf{v}}{\|\mathbf{v}\|} = \dfrac{a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z}{V}$.
- **Principal normal.** Since $\dot{s} = V$ is constant, $\ddot{s} = dV/dt = 0$ and the whole acceleration is normal: $\mathbf{a} = a_n \mathbf{e}_n = -a\omega^2 \mathbf{e}_R$. Taking the norm gives $a_n = a\omega^2$, and dividing the vector by it gives

$$\mathbf{e}_n = \frac{\mathbf{a}}{a_n} = \frac{-a\omega^2 \mathbf{e}_R}{a\omega^2} = -\mathbf{e}_R$$

The principal normal points radially inward, toward the axis of the cylinder.

- **Radius of curvature.** From $a_n = \dot{s}^2/\rho$:

$$\rho = \frac{V^2}{a_n} = \frac{a^2\omega^2 + v_z^2}{a\omega^2} = a + \frac{v_z^2}{a\omega^2}$$

- **Binormal.** Using $\mathbf{e}_R \times \mathbf{e}_\theta = \mathbf{e}_z$ and its cyclic permutations ($\mathbf{e}_z \times \mathbf{e}_R = \mathbf{e}_\theta$), together with the anticommutativity $\mathbf{e}_\theta \times \mathbf{e}_R = -\mathbf{e}_z$:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \frac{a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z}{V} \times (-\mathbf{e}_R) = \frac{-a\omega\,(\mathbf{e}_\theta \times \mathbf{e}_R) - v_z\,(\mathbf{e}_z \times \mathbf{e}_R)}{V} = \frac{-v_z\,\mathbf{e}_\theta + a\omega\,\mathbf{e}_z}{V}$$

- **Velocity and acceleration with s and ρ:**

$$\mathbf{v} = \dot{s}\,\mathbf{e}_t = V\,\mathbf{e}_t, \qquad \mathbf{a} = \ddot{s}\,\mathbf{e}_t + \frac{\dot{s}^2}{\rho}\,\mathbf{e}_n = \frac{V^2}{\rho}\,\mathbf{e}_n = a\omega^2\,\mathbf{e}_n$$

- **Second change-of-basis matrix.** Its columns are the components of $\mathbf{e}_t$, $\mathbf{e}_n$, $\mathbf{e}_b$ in $\mathcal{B}_{\text{cyl}}$:

$$[{}_{\text{cyl}}R_{\text{tnb}}] = \begin{pmatrix} 0 & -1 & 0 \\ a\omega/V & 0 & -v_z/V \\ v_z/V & 0 & a\omega/V \end{pmatrix}, \qquad RR^T = I$$

With $V^2 = a^2\omega^2 + v_z^2$, the rows have unit norm (1, $(a^2\omega^2 + v_z^2)/V^2 = 1$ and $(v_z^2 + a^2\omega^2)/V^2 = 1$) and are mutually orthogonal (row 1 against rows 2 and 3 gives 0, rows 2 and 3 give $a\omega v_z/V^2 - v_z a\omega/V^2 = 0$), so $RR^T = I$; expanding along the first row, $\det = -(-1)\left(\frac{a^2\omega^2}{V^2} + \frac{v_z^2}{V^2}\right) = 1$. Expanding the determinant along the first row (its only non-zero entry is -1 in position $(1,2)$, with cofactor sign -1):

$$\det = -(-1)\left[\frac{a\omega}{V}\cdot\frac{a\omega}{V} - \left(-\frac{v_z}{V}\right)\frac{v_z}{V}\right] = \frac{a^2\omega^2 + v_z^2}{V^2} = 1$$

The columns have unit norm and are mutually orthogonal, so $RR^T = I$; the composition $[{}_0R_{\text{tnb}}] = [{}_0R_{\text{cyl}}][{}_{\text{cyl}}R_{\text{tnb}}]$ is again a proper rotation (determinant $1 \cdot 1 = 1$).

3.2 Forces

Not applicable: kinematics-only problem. The trajectory is given, so no force enters the question.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The motion is prescribed by the statement, so there is no equation of motion to solve.

3.4 Integration and Answers

(e) Arc length as a function of time

> **Pedagogical Justification:** $s(t)$ measures the distance travelled along the trajectory from $t = 0$, i.e. the integral of the speed.

$$s(t) = \int_0^t v\,dt' = \int_0^t V\,dt' = V\,[t']_0^t = V(t-0) = t\sqrt{a^2\omega^2 + v_z^2}$$

(f) Trajectory as a function of the arc length

> **Pedagogical Justification:** inverting $s(t)$ gives $t(s) = s/V$, which is substituted into the parametric equations to remove the time:

$$x(s) = a\cos\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right), \qquad y(s) = a\sin\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right), \qquad z(s) = \frac{v_z s}{\sqrt{a^2\omega^2 + v_z^2}}$$

(g) Centre of curvature at $(a, 0, 0)$

> **Pedagogical Justification:** the centre C of the osculating circle lies on the principal normal at distance ρ: $\mathbf{r}_C = \mathbf{r}_P + \rho\,\mathbf{e}_n$ (Notes Sec. 2.4).

- The helix passes through $(a, 0, 0)$ where $z = v_z t = 0$, i.e. at $t = 0$. The statement says $t > 0$, so we take the motion law to hold for $t \geq 0$ (the point is the limit $t \to 0^+$; all formulas are continuous there). At that point $\mathbf{r}_P(0) = a\,\mathbf{i}_0$ and $\mathbf{e}_n(0) = -\mathbf{e}_R(0) = -\mathbf{i}_0$:

$$\mathbf{r}_C = a\,\mathbf{i}_0 + \left(a + \frac{v_z^2}{a\omega^2}\right)(-\mathbf{i}_0) = -\frac{v_z^2}{a\omega^2}\,\mathbf{i}_0, \qquad \boxed{C = \left(-\frac{v_z^2}{a\omega^2},\ 0,\ 0\right)}$$

Interpretation & Checks

- **Dimensional check:** $[a\omega] = \mathrm{m} \cdot \mathrm{s}^{-1}$ and $[v_z] = \mathrm{m/s}$, so V is in m/s. The offset $\left[\dfrac{v_z^2}{a\omega^2}\right] = \dfrac{(\mathrm{m/s})^2}{\mathrm{m} \cdot \mathrm{s}^{-2}} = \mathrm{m}$.
- **Limiting cases:** if $v_z \to 0$ (planar circle) $\rho = a$ and $C = (0, 0, 0)$, the centre of the circle. If $\omega \to 0$ (vertical ascent) $\rho \to \infty$: the curvature vanishes and the trajectory is a straight line.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 1 in its solution keys. The results are verified by independent derivation: sympy reproduces $\|\mathbf{v}\| = V$, $a_n = a\omega^2$, $\rho = a + v_z^2/(a\omega^2)$ (also from $\rho = \|\mathbf{v}\|^3/\|\mathbf{v} \times \mathbf{a}\|$), the binormal, the centre of curvature and $\det[{}_{\mathrm{cyl}}R_{\mathrm{tnb}}] = 1$ with $RR^T = I$.

Source: Problems.pdf (Taken from MIT OCW)